EE3032 Introduction to VLSI Design—Homework 2

1. **Non-ideal Effect** (5 points)
   Fig. 1 shows a simple circuit with a capacitance. Assume that the threshold voltage and the pn-junction leakage current of the NMOS transistor are V_t and I_L, respectively. Also, the switching threshold voltage of the buffer is $0.4V_{dd}$. Complete the following questions.

   (a) If we turn on the NMOS by asserting SL=1 and precharging the capacitance through the PMOS transistor by asserting Pre=0, what voltage level of the capacitance can be precharged to? (2 point)

   (b) If we turn off the NMOS and PMOS, how long the logic 1 (output still can read data 1) can be retained? (3 points)

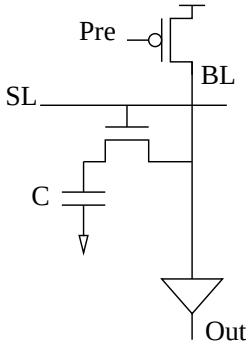

Figure 1: A simple circuit with a capacitance.

2. **Logic Gate** (5 points)
   Fig. 2 shows a CMOS logic gate. If the size of NMOS and PMOS transistors is W/L, the corresponding turned-on resistance is R_n and R_p, respectively, regardless of the transistor is operated at linear or saturation regions. Also, $R_p = 2R_n$, where $\mu_n = 2\mu_p$ is assumed. Complete the following questions.

   (a) What is the Boolean function realized by the logic gate? (1 points)

   (b) If the smallest voltage of logic 0 is $0.2V_{dd}$, determine the sizes of the transistors such that the logic gate can operate correctly. (4 points)

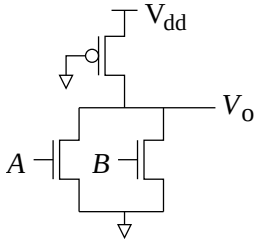

Figure 2: CMOS logic gate